

David Natsvlishvili

Senior Software Engineer

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Bio:

AI engineer with 16 years of experience designing, scaling, and deploying **machine learning** and **generative AI systems** across defense, enterprise, and global consumer platforms. Currently a **Senior AI Engineer at Meta**, driving large-scale initiatives in **LLaMA**, **RAG**, **multimodal embeddings**, and **PyTorch-based distributed training**. Led the development of **internal AI platforms** that enable billions of daily predictions and delivered **full-stack GenAI tools** that allow product teams to experiment with LLMs in production workflows.

Previously at **Sutherland**, built **predictive analytics** and **NLP solutions** for enterprise clients in **finance and telecom**, deploying **fraud detection**, **churn prediction**, and **intent classification systems** into production environments via **APIs** and **Java services**. Began career at **Leidos** as a **Data Scientist**, applying **statistical modeling**, **anomaly detection**, and **early ML techniques** for defense and government programs while developing **ETL pipelines** and **C#/.NET dashboards** that made model outputs accessible to analysts.

Recognized for bridging **AI research and engineering execution**, with deep expertise in **PyTorch**, **TensorFlow**, **FAISS**, **Kubernetes**, and modern **MLOps practices**. Adept at mentoring teams, scaling infrastructure, and translating cutting-edge AI into **measurable business impact**.

Education:

University: **National Louis University**

Chicago, Illinois, USA

Degree: Bachelor's Degree, Computer Science

Jun 2004 – May 2008

Professional Experience:

Meta

Senior AI Engineer

Apr 2019 – Present

- Drove applied AI initiatives within FAIR and product engineering, focusing on PyTorch-based large-scale model training and deployment to support billions of daily predictions.
- Built a retrieval-augmented generation (RAG) platform on top of FAISS + PyTorch Lightning, powering internal semantic search across Meta's massive knowledge bases.
- Prototyped and shipped a GenAI experimentation tool using LLaMA models with a React + GraphQL frontend and Python microservices backend — gave product teams a way to plug LLMs directly into user workflows.
- Improved distributed training and inference workflows on Kubernetes and Ray, making large-scale PyTorch jobs more reliable and faster to ship.
- Partnered with infra and product teams to scale multimodal embeddings (vision + text) for recommendation and ranking systems.
- Established MLOps best practices across teams — drift detection, MLflow-based experiment tracking, and custom Grafana dashboards for production LLMs.
- Mentored engineers and shared best practices in internal PyTorch guilds and AI infra forums, accelerating adoption of LLaMA, RAG, and embedding search across teams.

Sutherland

Machine Learning Engineer

Feb 2012 – Apr 2019

- Designed and implemented end-to-end ML solutions for global enterprise clients in finance, telecom, and customer support, with a focus on predictive analytics, NLP, and anomaly detection.
- Developed and deployed text classification, sentiment analysis, and intent detection models using scikit-learn, spaCy, and TensorFlow, significantly reducing manual case handling.
- Built fraud detection and churn prediction systems leveraging time-series forecasting, GLMs, and gradient boosting — improved detection accuracy and reduced false positives.
- Partnered with engineering teams to productionize ML models through REST APIs, Java-based services, and SQL pipelines, ensuring seamless integration with client applications.
- Established feature engineering and evaluation frameworks (cross-validation, cohort analysis, residual tracking) to improve model robustness and reliability in production.

- Collaborated with cross-functional stakeholders to translate raw data into scalable AI-driven solutions, delivering measurable cost savings and operational efficiencies.
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Leidos

Data Scientist

Jun 2008 – Feb 2012

- Developed statistical and machine learning models for defense and government programs, focusing on predictive analytics, anomaly detection, and risk assessment.
 - Designed and maintained ETL pipelines in Python and SQL to consolidate and process large, heterogeneous datasets for downstream analysis.
 - Applied classical ML techniques (logistic regression, clustering, ensemble methods) to improve intelligence analysis and operational decision-making.
 - Built interactive visualization and reporting tools using C#/.NET, enabling analysts to explore model outputs and trends in real time.
 - Partnered with engineering teams to integrate models into production workflows, balancing accuracy, interpretability, and computational efficiency.
 - Advocated for the early adoption of data-driven approaches, helping teams transition from manual reporting toward automated, AI-enabled insights.
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Tech Stack

AI / ML: PyTorch, TensorFlow, PyTorch Lightning, scikit-learn, spaCy, Hugging Face Transformers, LLaMA, RAG pipelines, FAISS, pgvector, Pinecone, Ray

MLOps / Data: MLflow, Airflow, Kubeflow, Prefect, SQL, BigQuery, Snowflake, Spark, ETL pipelines, feature engineering, model monitoring

Infrastructure: Kubernetes, Docker, Horizon, AWS, GCP, Azure, Prometheus, Grafana, CI/CD (GitHub Actions, GitLab, Jenkins)

Full Stack / Backend: Python, Go, Java, C#/.NET, REST APIs, GraphQL, FastAPI, Flask, Node.js, microservices architecture

Frontend / Tools: React, Next.js, JavaScript/TypeScript, HTML/CSS, dashboard/reporting development

Other: NLP, computer vision, multimodal embeddings, predictive modeling, anomaly detection, time-series forecasting, statistical analysis